Philip S. Chodrow

Assistant Professor of Computer Science
Middlebury College

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Experience

Assistant Professor of Computer Science — *2022 – present*
Middlebury College — *Middlebury, VT*

Hedrick Visiting Assistant Adjunct Professor of Mathematics — *2020 – 2022*
University of California, Los Angeles — *Los Angeles, CA*

Education

PhD in Operations Research — *2015-2020*
Massachusetts Institute of Technology — *Cambridge, MA*

BA in Mathematics and Philosophy — *2008-2012*
Swarthmore College — *Swarthmore, PA*

Grants, Fellowships and Funding

RUI: Network Evolution with Unobserved Mechanisms ($287,522) — *2024-2027*
National Science Foundation (NSF) — *Middlebury, VT*

Graduate Research Fellowship ($138,000) — *2017-2020*
National Science Foundation (NSF) — *Cambridge, MA*

Fulbright Research Award (~$28,000) — *2012-2013*
US-Norway Fulbright Foundation — *Oslo, Norway*

Papers

18. Violet Ross°, Francis Cataldo° and **PSC** (2026): Growing Hypergraphs with Homophily. *arXiv:2607.16046.* (°Undergraduate mentees)
17. **PSC** (2025): Equivalence of Informations Characterizes Bregman Divergences. *Entropy* 27, 766.
16. Xie He, **PSC** and Peter J. Mucha (2025): Edge Correlations and Link Prediction in Growing Hypergraphs. *Physical Review E* 112, 24305. (two equal first authors)
15. **PSC**, Harlin Lee, Natalie Lao and Vincent Monardo (2025): "Active Learning for Reducing Gender Gaps in Undergraduate Computing and Data Science," in *Advances in Data Science: Women in Data Science and Mathematics (WiSDM) 2023* Association for Women in Mathematics Series. (Springer), 362.
14. Heather Z. Brooks, **PSC** and Mason A. Porter (2024): Emergence of Polarization in a Sigmoidal Bounded-Confidence Model of Opinion Dynamics. *SIAM Journal on Applied Dynamical Systems* 23, 1442–1470.

13. Jürgen Kritschgau, Daniel Kaiser, Oliver Alvarado Rodriguez, Ilya Amburg, Jessalyn Bolkema, Thomas Grubb, Fangfei Lan, Sepideh Maleki, **PSC** and Bill Kay (2024): Community Detection in Hypergraphs via Mutual Information Maximization. *Scientific Reports* 14, 6933.

12. Quindel Jones, Andrés R. Vindas-Meléndez, Ariana Mendible, Manuchehr Aminian, Heather Z. Brooks, Nathan Alexander, Carrie Diaz Eaton and **PSC** (2023): Data Science and Social Justice in the Mathematics Community. *Notices of the American Mathematical Society* 70, 1.

11. **PSC**, Nicole Eikmeier and Jamie L. Haddock (2023): Nonbacktracking Spectral Clustering of Nonuniform Hypergraphs. *SIAM Journal on Mathematics of Data Science* 5, 251–279.

10. Mary Angelique G. Demetillo, Colin Harkins, Brian C. McDonald, **PSC**, Kang Sun and Sally E. Pusede (2021): Space-based Observational Constraints on NO2 Air Pollution Inequality from Diesel Traffic in Major US Cities. *Geophysical Research Letters* 48, e2021GL094333.

9. **PSC**, Nate Veldt and Austin R. Benson (2021): Generative Hypergraph Clustering: From Blockmodels to Modularity. *Science Advances* 7, eabh1303.

8. Mari Kawakatsu, **PSC**, Nicole Eikmeier and Daniel B. Larremore (2021): Emergence of Hierarchy in Networked Endorsement Dynamics. *Proceedings of the National Academy of Sciences* 118, e2015188118. (three equal first authors)

7. **PSC** (2020): Configuration Models of Random Hypergraphs. *Journal of Complex Networks* 8, cnaa018.

6. **PSC** (2020): Moments of Uniform Random Multigraphs with Fixed Degree Sequences. *SIAM Journal on Mathematics of Data Science* 2, 1034–1065.

5. **PSC** and Peter J. Mucha (2020): Local Symmetry and Global Structure in Adaptive Voter Models. *SIAM Journal on Applied Mathematics* 80, 620–638.

4. **PSC** and Andrew Mellor (2020): Annotated Hypergraphs: Models and Applications. *Applied Network Science* 5, 9.

3. **PSC** (2017): Structure and Information in Spatial Segregation. *Proceedings of the National Academy of Sciences* 114, 11591–11596.

2. **PSC**, Zeyad Al-Awwad, Shan Jiang and Marta C. González (2016): Demand and Congestion in Multiplex Transportation Networks. *PloS ONE* 11, e0161738.

1. **PSC**, Cole Franks and Brian Lins (2013): Upper and Lower Bounds for the Iterates of Order-Preserving Homogeneous Maps on Cones. *Linear Algebra and its Applications* 439, 999–1005.

Other Scholarly Writings

4. Maximino Aldana, Roni Barak Ventura, Heather Z. Brooks, **PSC**, Filipe Georgiou, Joseph Johnson, Krešimir Josić, Zachary P. Kilpatrick, Kath Landgren, Andrew Nugent, Maurizio Porfiri, Nancy Rodriguez, et al. (2026): Modeling Social Systems: Transparency, Reproducibility, and Responsibility. *SIAM News*.

3. **PSC**, Manuchehr Aminian, Carlos Paniagua and Jude Higdon (2021): Persons Charged with Violations of 21 U.S.C. §846: Poverty, Unemployment, Education, and Sentences.

2. **PSC**, Mari Kawakatsu, Nicole Eikmeier and Daniel B. Larremore (2021): Networks, Dynamics, and Prestige: How Hierarchies Emerge from Individual Choices. *SIAM News Blog*.

1. **PSC** (2017): Divergence, Entropy, Information: An Opinionated Introduction to Information Theory. *arXiv: 1708.07459*.

Invited Talks

Dynamics of Gender Representation in Professional Mathematics

- 2026 CU Boulder
- 2025 UVM Complex Systems Institute
- 2025 CMO BIRS: Collective Social Phenomena
- 2025 SIAM Conference on Applied Dynamical Systems
- 2025 Michigan State University CMSE Seminar
- 2024 Tufts University Applied Mathematics Seminar

Teaching Network Science with Interactive Lecture Notes

- 2026 NetSci Ed at NetSci

Connection, Computation, and Complex Systems in Undergraduate Education

- 2024 Conference on Network Science (NetSci)

Edge-Correlated Growing Hypergraphs

- 2026 TopoNets at NetsSci
- 2025 CIRRELT Network Science Workshop
- 2024 BRAINS Seminar: Bridging AI, Networks and Hypergraphs

Challenges and Opportunities in Hypergraph Network Data Science

- 2023 Vermont-KIAS Workshop on Group Interactions in Network Science
- 2023 Joint Math Meetings: Special Session on Hypergraph Data Science

Feedback Loops from Ranking in Networks

- 2025 Complex Networks Winter Workshop
- 2023 Smith College Mathematics Seminar
- 2021 REU on Kaczmarz Methods for Large-Scale Data Analysis, Claremont Colleges
- 2021 UCLA Undergraduate Mathematics Student Association

Generative Hypergraph Clustering: Scalable Heuristics and Sparse Thresholds

- 2024 Toronto Metropolitan University Network Science Seminar
- 2023 University of Vermont Joint Lab Seminar
- 2023 University of Edinburgh Informatics Seminar
- 2023 University of Glasgow Dept.Computer Science Seminar
- 2022 Oxford University Networks Seminar
- 2022 Swarthmore College Mathematics Seminar
- 2022 Middlebury College Computer Science Seminar
- 2022 Colgate University Computer Science Seminar
- 2022 Wesleyan Computer Science Seminar
- 2021 Canadian Mathematical Society Winter Meeting
- 2021 Claremont Center for the Mathematical Sciences
- 2021 UCLA Undergraduate Mathematics Student Association
- 2021 Grinnell Computer Science Seminar
- 2021 Conference on Network Science (NetSci): Satellite on Higher-Order Models in Network Science
- 2021 Workshop on Graphs and Complex Structures for Learning and Reasoning at (AAAI)

Public Talks: Data Science and AI

2023 Middlebury DLINQ Summer Workshop
2023 Middlebury TEDx
2023 Middlebury Faculty At Home

Random Graphs for Data Science

2020 UCLA Network Science Seminar
2020 MIT ORC Student Seminar
2019 Northeastern University Network Science Seminar

Other Professional Events

Workshops

2025 Complex Networks Winter Workshop
Quebec City, CA

2025 Collective Social Phenomena: Dynamics and Data
Casa Matemática Oaxaca (CMO)

2023 Data Science and Social Justice: Networks, Policy, and Education
ICERM

2023 AMS Mathemematics Research Community: Mathematics of Complex Social Systems

2022 Data Science and Social Justice: Networks, Policy, and Education
ICERM

2022 AMS Mathemematics Research Community: Models and Methods for Sparse (Hyper)Network Science

2019 Complex Networks Winter Workshop
Quebec City, CA

Invited Panels

2022 Panel on Academic Job Interviews
Pomona REU

2022 Data Science and the Social Sciences
UCLA Data Science Union

2021 Early-Career Research Panel
Northeastern Regional Conference on Complex Systems

2021 Perspectives on Graphs and Complex Structures for Learning and Reasoning
AAAI

Recognition

Finalist Nominee, Inclusive Design for Learning Award *2025*
Middlebury CTLR and Advisory Group on Disability, Access, and Inclusion *Middlebury, VT*

"...highlights an instructor's embrace of Inclusive Design and the role of Inclusive Design in high quality instruction."

Liggett Instructor Distinguished Faculty Teaching Award *2022*
Department of Mathematics, UCLA *Los Angeles, CA*

"Recognizing excellence in contribution to the instructional mission of the Department."

Ivy Award *2012*
Swarthmore College *Swarthmore, PA*

"Recognizing the man of the graduating class who is outstanding in leadership, scholarship, and contributions to the Swarthmore College community."

Travel and Conference Awards

2022 SIAM Travel Award, *SIAM Conference on Mathematics of Data Science*

2021 SIAM Travel Award, *SIAM Conference on Applications of Dynamical Systems*

2020 Best Poster Award, *Northeastern Regional Conference on Complex Systems*

Pedagogical Writings

Lecture Notes on Machine Learning *2026*
https://www.philchodrow.prof/ml-notes-update/ *Online*

Lecture Notes on Network Science (with Heather Brooks) *2025*
https://network-science-notes.github.io/ *Online*

Lecture Notes on Applied Data Science (with Michael Linderman) *2026*
https://middcs.github.io/data-science-notes/ *Online*

Teaching

Middlebury College

S25	CSCI 0442: *Network Science*
S23×2, S24, S25, S26×2	CSCI 0451: *Machine Learning*
W26	CSCI 1010: *Applied Data Science*
F23×2, F24×2, F25×2	CSCI 0200: *Mathematical Foundations of Computing*
W24, S24, F24, W25	CSCI 0702: *Senior Thesis*
F24	CSCI 0500: *Advanced Study*
F22×2	CSCI 0145: *Introduction to Computing*

University of California, Los Angeles

S22	MATH 168: *Introduction to Networks*
S21, F21, W22	PIC 16B: *Python Programming with Applications II*
F20, W21, S21, F21	PIC 16A: *Python Programming with Applications I*

Thesis and Research Students

PhD Committees

2027 William Thompson Complex Systems, University of Vermont

Middlebury Thesis Advisees

2025 Francis Cataldo *Community Detection in Growing Hypergraphs*

2025 Violet Ross *Growing Hypergraphs with Node Attributes*

2024	Bell Luo	*Efficient Sampling from Hypergraph Configuration Models*
	Jamie Hackney	*Opinion Dynamics and Electoral Systems*

Other Middlebury Research Students

2024	Justin Corke	*Degree Assortativity of Union Hypergraphs*

Middlebury Thesis Committees

2026	Azan Umman	(*Physics and Astronomy*)
2025	Alex Takoudes	(*Earth and Climate Science*)
	Liam Patty	(*Computer Science*)
	Alexis Ballo	(*Computer Science*)
2024	Toby Pouler	(*Computer Science*)
	Otis Milliken	(*Computer Science*)
2023	Ben Rahill	(*Computer Science*)

Professional Development

2024 – 2025	Cultural Competency in Computing (3C) Fellowship
2023	Middlebury College Engaged Listening Project
2021	MAA Project NeXT
2020	UCLA CEILS Introduction to Evidence-Based Pedagogy

Professional Service

Scholarly Reviews

SIAM Review	*SIAM J. Math. Data Sci.*	*SIAM J. Appl. Dyn. Sys.*
SIAM J. Matrix Analysis	*Nature Communications*	*Science Advances*
Proc. Nat. Acad. Sci. (PNAS)	*J. Appl. & Comp. Topology*	*J. Machine Learning Research*
Physical Review Letters	*Physical Review X*	*Physical Review E*
Discrete Mathematics	*IEEE Trans. Network Sci. & Eng.*	*Nature Human Behavior*
EPJ Data Science	*Social Network Analysis & Mining*	*Networks & Spatial Economics*
Chaos	*Int'l J. Geo. Info. Sys.*	*PLoS One*

Grant Review Panels

Applied and Computational Mathematics — *2025*
NSF Division of Mathematical Sciences (DMS) — *Online*

Applied and Computational Mathematics — *2022*

NSF Division of Mathematical Sciences (DMS) *Online*

Events Co-Organized

A Blue Start: Higher Order Hackathon at NetSci *2026*
Conference on Network Science (NetSci) *Boston, MA*

Minisymposium on Mathematics of Network Science *2025*
SIAM Annual Meeting *Montreal, QC*

Minisymposium on Data, Inference, and Dynamics in Complex Social Systems *2025*
SIAM Conference on Applied Dynamical Systems *Denver, CO*

Minisymposium on Tensor Methods in Network Science *2022*
SIAM Conference on Mathematics of Data Science *San Diego, CA*

Southern California Applied Mathematics Symposium (SoCAMS) *2022*
Harvey Mudd College *Claremont, CA*

MAA Project NExT Workshop on Projects Across the Mathematics Curriculum *2022*
Joint Math Meetings *Online*

Workshop on Graphs and Complex Structures for Learning and Reasoning *2022*
Association for the Advancement of Artificial Intelligence (AAAI) *Vancouver, CA*

Other Professional Service

Poster Session Judge *2025*
SIAM Conference on Applied Dynamical Systems *Denver, CO*

Professional Societies

Society for Industrial and Applied Mathematics (SIAM)
Mathematical Association of America (MAA)
Association for Women in Mathematics (AWM)
National Association of Mathematicians (NAM)

Institutional Service

Committee Membership

Technology Advisory Committee *2025-*
Middlebury College *Middlebury, VT*

Ad Hoc Committee on Curricular Requirements *2024-2025*
Middlebury College *Middlebury, VT*

Steering Committee of Midd.Data *2023-*
Middlebury College *Middlebury, VT*